

Hero Adjusted Plus–Minus in *Marvel Rivals*

Intention-to-treat estimates from 477,483 competitive matches

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If all we wanted to know was which heroes win the most games, there would be no reason to build this model. We could sort the leaderboard by raw win rate, declare the hero at the top broken, declare the hero at the bottom useless, and be done by lunch. The problem is that a hero does not randomly materialise into an otherwise identical match. Better players choose some heroes more often. Some heroes appear disproportionately in strong or weak compositions. Some are tied to particular team-ups. Maps differ. Teams differ in skill. Role structures differ. A 55% raw win rate can therefore mean that a hero is strong, that the situations in which the hero gets picked are strong, or some mixture of the two, and the raw number cannot tell you which.

The object of this report is narrower and more useful. Using 477,483 competitive PC ranked matches from Season 9.5, I estimate the association between starting a particular hero and winning after holding fixed the observable features of the match that can reasonably be separated from hero identity. The resulting number is an adjusted plus–minus, or APM. It answers the question: if an average hero of the same role is replaced by this hero at the start of an otherwise comparable match, how much does the predicted probability of winning change? That is not the same question as raw win rate, and pretending that it is would defeat most of the purpose of having 477,483 matches in the first place.

The first nuisance is team composition. *Marvel Rivals* does not force every team into the same role structure, and those structures have very different outcomes before anyone asks which specific heroes occupy them. The standard composition of two Tanks, two Damage heroes, and two Supports accounts for 75% of teams in the sample and wins 53.5% of its matches. A 1–3–2 team is not the same object. Neither is 2–1–3, 0–4–2, or any of the more exotic things ranked players occasionally decide are a good idea. In the data, one Tank with three Damage heroes wins 42.2% of its matches, and two Tanks with three Supports wins 40.4%. If composition were ignored, the regression would happily blame individual heroes for losses partly caused by the structure around them. That would be convenient. It would also be wrong.

I therefore control for composition directly. Every Tank–Damage–Support combination that appears at least 100 times receives its own indicator, entered as the difference between the two teams, and the rare remainder is pooled into a single “other” category. The point is simple. The composition block gets to explain whether a strange role structure is good or bad. The hero block gets to explain which heroes are relatively good or bad within their roles. Those are different questions, so the model should not force one coefficient to answer both.

Each match is one observation, and the outcome is whether side 0 wins. A hero enters as a signed contrast between the two teams. If Hela starts on side 0, that adds one to the Hela regressor. If she starts on side 1, it subtracts one. The hero coefficients are constrained to sum to zero separately within Tanks, Damage heroes, and Supports, so The Thing is compared with the average Tank, Black Cat with the average Damage hero, and Mantis with the average Support. Cross-role comparisons are deliberately not the object being estimated. The rest of the specification absorbs the obvious sources of variation. Each of the 16 maps receives its own intercept, the pre-match rank-score difference between the teams enters directly, and 100 designated team-up contrasts sit alongside the composition contrasts just described. The model is an unpenalised logistic regression with HC1 heteroskedasticity-robust standard errors.

The annoying question: who gets credit when someone swaps?

Hero attribution is where an apparently innocent modelling decision becomes much less innocent. Suppose a player starts Hela, plays her for two minutes, swaps to Punisher, and spends the remaining eight minutes there. There are at least three obvious ways to encode that player. Under starting-hero attribution (W0), the match counts as 100% Hela, and the question being asked is simply which hero the player started on. Under most-played-hero attribution (W2), the same match counts as 100% Punisher, because Punisher received most of the player's minutes. Under play-time weighting (W1), the observation becomes 20% Hela and 80% Punisher.

Play-time weighting initially sounds like the sophisticated choice. The data record the exact time spent on each hero, so why throw information away? Unfortunately, the extra information is produced during the match. Players do not swap at random. They swap because they are losing a matchup, because the enemy composition changed, because a hero is not working, or because the match itself has revealed something that was unavailable at the start. Weighting by realised play time therefore lets the outcome-generating process help construct the explanatory variable. The model starts using the match to explain the match. This produces wonderfully impressive coefficients, which is precisely the problem.

Figure 1 makes the distinction concrete and then shows what it does to the estimates. Moving from the least outcome-dependent rule to the most outcome-dependent one, the cross-hero standard deviation of the 55 APM estimates rises from 2.60 percentage points under starting-hero attribution to 4.27 under most-played-hero attribution and 6.38 under play-time weighting. Magik is the extreme case, moving from +3.47 points to +7.94 and then +14.57 across the three rules. The interesting part is that the rankings do not collapse. Their rank correlations with the starting-hero ranking are still 0.937 for the most-played rule and 0.890 for play-time weighting. What changes is the magnitude. The more information from after the opening seconds is fed into attribution, the larger the hero effects become, and that is exactly the pattern to worry about if the additional accuracy is partly post-treatment information leaking into the regressors.

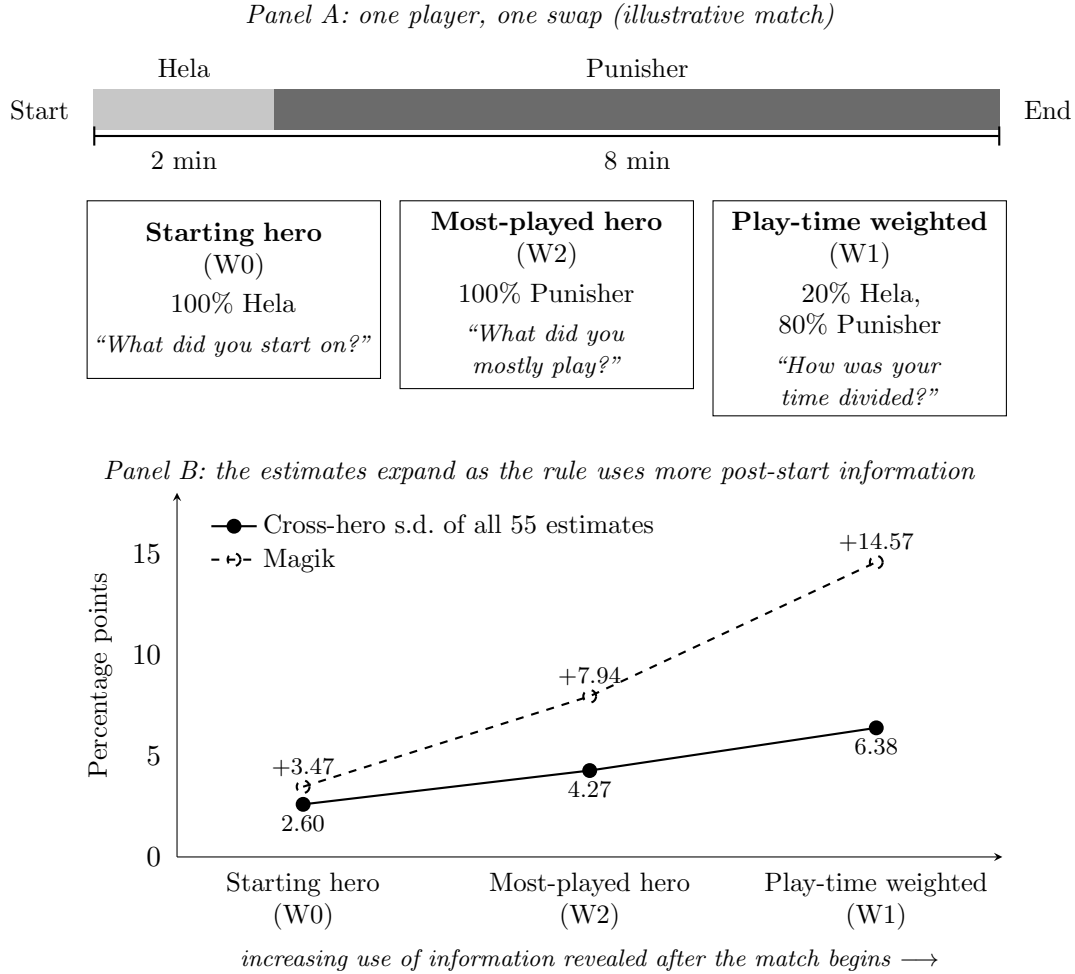
For the headline results I therefore use starting-hero attribution. The API records heroes chronologically by first appearance, so the first hero gives an assignment fixed before the rest of the match is known. If someone starts a hero and abandons it thirty seconds later, that hero still receives the match. This unquestionably attenuates some effects. A hero that players frequently abandon will look less extreme than it does under play-time weighting. But that behaviour is not irrelevant to the hero's practical value, and the rule buys a much cleaner interpretation: what happens after a player begins the match on this hero? That is an intention-to-treat object rather than an attempt to reconstruct, after the fact, which hero deserves the result.

Why the comparison is within role

There is a second identification issue that is easy to miss, because the regression returns numbers whether or not it is handled. The number of Tanks, Damage heroes, and Supports on a team is itself part of the composition. If all hero coefficients are placed under one global normalisation, the hero block and the composition block share an almost exactly collinear direction corresponding to role counts. The model is then asked to decide whether having more Tanks matters because of Tank hero effects or because the team chose a different role structure. There is no good reason to make it arbitrate a distinction that can be imposed directly.

The clean solution is to normalise within each role. Tank coefficients sum to zero across Tanks, Damage coefficients across Damage heroes, and Support coefficients across Supports. Composition effects are then left to the composition controls, and hero effects measure variation among heroes filling the same strategic slot. Re-estimating the model under this parameterisation barely changes the substantive ordering. The rank correlation with the earlier estimates is 0.9999 and the mean absolute change in a

Figure 1: Who gets credit when a player swaps heroes, and what that choice does to the estimates



Notes. Panel A is a stylised match. The three rules agree about any player who never swaps and disagree only about players who do, which is where the outcome of the match can leak into the regressors. Panel B plots, for each rule, the cross-hero standard deviation of the 55 within-role estimates (filled circles), all fitted on the same 477,483 matches, and the estimate for Magik (open circles), the hero whose estimate moves most between the starting-hero and play-time-weighted rules. Appendix Table A3 lists every hero under every rule.

coefficient is 0.010 percentage points. Fixing the identification problem does not manufacture the result. It makes the interpretation defensible.

What the model finds

There is substantial dispersion within all three roles (Table 1). Among Tanks, The Thing has the largest positive adjusted effect at 5.46 percentage points relative to the average Tank, followed by Magneto, Angela, Peni Parker, Hulk, and Thor. Among Damage heroes, Black Cat leads at 4.85 points, followed by Magik, Storm, Daredevil, and Mister Fantastic. Among Supports, Mantis is the clear outlier at 6.76 points, followed by Rocket Raccoon, Ultron, Adam Warlock, and Loki.

These numbers are marginal effects evaluated at a balanced match. The Thing’s 5.46 does not mean that every team picking The Thing raises its observed win rate by 5.46 points. It means that, at the evaluation point, replacing an average Tank with The Thing is associated with about a 5.46 percentage-point increase in predicted win probability after conditioning on the controls in the model. 10

of the 55 heroes cannot be statistically distinguished from their role average after Benjamini–Hochberg false-discovery-rate correction.

Table 1 also shows why raw win rate and adjusted value should not be treated as synonyms. Alongside the APM and its uncertainty it reports starts, pick share within the crawl, the raw play-time-weighted win rate, and the change produced by replacing starting-hero attribution with play-time weighting. Some heroes move substantially once the surrounding match is controlled for. Others move substantially when swaps are allowed to influence attribution. Those are different sources of distortion, and the table exposes both rather than pretending that one number is the hero’s true strength.

Does any of this survive outside the sample?

A large regression can always become very good at describing the data used to fit it, which is not especially interesting. The more useful test is whether it predicts matches it has not seen. I therefore fit the model on the 437,453 matches played before the final 7 days of the sample and evaluate it on the remaining 40,030. The split is chronological rather than random, so the holdout is genuinely later data and the meta drift that a random split would smuggle across the boundary stays on the test side.

Figure 2 evaluates the model in two ways. Panel A groups holdout team-sides by the hero they started and compares each group’s realised win rate with the model’s average predicted win probability. If the model says that teams starting a given hero should win 53% of the time, those teams should win approximately 53% of the time in data the model never saw. They largely do. Across the 55 heroes, predicted and realised holdout win rates have a correlation of 0.957 and a mean absolute gap of 0.50 percentage points. More revealingly, if the hero terms are removed while map, rank score, composition, and team-ups are retained, the correlation falls to 0.139. The remaining controls know something about the environments different heroes appear in, but they do not reproduce the cross-hero outcome pattern on their own. Panel B repeats the exercise for composition shapes and produces a correlation of 0.975 with a mean absolute gap of 1.19 points.

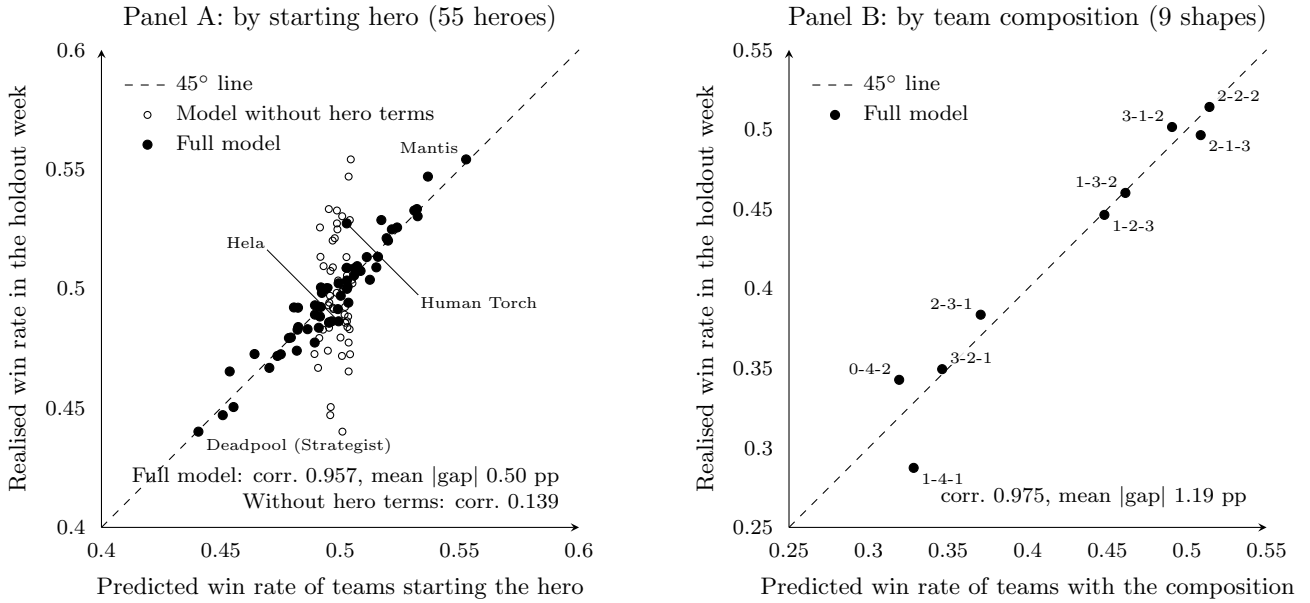
The model is also well calibrated at the match level (Appendix Figure A1). Its holdout calibration slope is 1.067 with a standard error of 0.025. A slope of one would be ideal, and this one is slightly above it, so the model is, if anything, mildly under-confident rather than over-confident. Holdout log loss is 0.6674 against 0.6928 for a constant prediction, and the area under the ROC curve is 0.627. Adding the hero block to a specification that already includes map, rank score, and composition reduces holdout log loss from 0.6819 to 0.6698, and adding team-ups takes it to 0.6674 (Appendix Table A2). The hero terms are therefore not merely producing attractive retrospective rankings. They improve prediction on matches played later.

Several specification checks point the same way (Appendix Table A1). Permuting hero assignments against outcomes destroys 97.4% of the measured signal. The within-role restrictions hold to machine precision. Correcting the role-composition identification issue raises the rank correlation between APM and raw hero win rate from 0.671 to 0.801, while the within-role reparameterisation itself barely moves the estimates. Map-specific intercepts are retained because side-0 win rates vary from 49.42% to 52.52% across the 16 maps. None of these checks proves that the coefficients are causal. They do make the simpler story, that the regression generates hero rankings by accident, increasingly difficult to maintain.

Team-ups are an extra effect, not the whole pair

The team-up coefficients (Appendix Table A4) estimate a different object from the hero coefficients. They ask whether a designated pair performs better or worse than the two heroes’ individual effects would predict. Squirrel Girl with Iron Man carries an estimated 8.66 percentage-point premium, Blade with Moon Knight 8.30, Rogue with Gambit 6.95, and Adam Warlock with Ultron 6.82. These values are not the total strength of the pairs. If two individually excellent heroes also have a positive team-up coefficient, their combined contribution is both hero terms plus the premium. Reading 8.66 as the gain

Figure 2: Does the model predict outcomes it has not seen? Holdout week, by hero and by composition



Notes. The model is fitted on the 437,453 matches before the final 7 days and evaluated on the 40,030 matches of that week. Each holdout match contributes two team-sides. Panel A groups team-sides by each hero they started: the horizontal axis is the model's mean predicted win probability for those team-sides, the vertical axis the share that actually won. Filled circles use the full model. Open circles use the same model with the hero contrasts removed, so their horizontal spread is only what map, rank score, composition, and team-ups can explain about a hero's teams. Labels mark the two heroes farthest from the line and the two extremes. Panel B groups team-sides by composition shape (Tanks–Damage–Supports), showing every shape with at least 100 holdout team-sides. Appendix Figure A1 gives the match-level calibration.

from fielding Squirrel Girl and Iron Man together counts the wrong object. The coefficient measures what the pair adds beyond what the two heroes already explain separately.

What this model does not prove

There are several reasons not to turn the coefficient table into scripture. First, the exact magnitudes depend noticeably on how swaps are attributed. Under starting-hero attribution the cross-hero standard deviation is 2.60 points. It rises to 4.27 under most-played-hero attribution and 6.38 under play-time weighting, while the rankings retain correlations of 0.937 and 0.890 with the starting-hero ranking. The ordering is therefore considerably more robust than the size of the gaps. If the question is which heroes consistently look strong or weak, the model has a stable answer. If the question is whether a hero is worth exactly +3.47 percentage points, much more caution is warranted.

Second, these are adjusted associations, not randomised treatment effects. Hero selection is endogenous. Rank score controls for general differences in player strength, but it does not say whether a player has five hundred hours on Magik or picked the hero because someone in chat said it was S-tier. Hero-specific proficiency can therefore remain correlated with both hero selection and outcomes. Player-by-hero experience would be the obvious control, and it is not in the current specification. The regression reduces several major sources of confounding. It does not abolish selection by decree.

Third, the confidence intervals are probably too narrow. HC1 treats matches as the primary observations, but players recur across matches and each match contains twelve of them, so the true dependence structure is overlapping and cross-classified. A player-cluster bootstrap would address this properly, but the implementation as specified would need roughly 656 hours of computation at the

current sample size. This does not change the point estimates. It does mean the uncertainty around them should be read more conservatively than the HC1 intervals suggest.

Finally, the tails of the data are imperfect. 18 composition shapes receive their own indicators, while 9 extremely rare shapes, together only 0.024% of team instances, share the “other” category and one coefficient. The crawl is also not a random sample of the ranked population. Profile privacy rises sharply at high rank scores and reaches roughly 63% above 5,000, and because the crawl expands only through public profiles, the very highest-ranked players are structurally under-sampled. The sample is enormous. It is not omniscient.

What I would take from it

The safest interpretation is also the useful one. These estimates measure how starting a hero is associated with winning after accounting for observed differences in map, team strength, role structure, and designated team-ups. They are most naturally read within role. The rankings are substantially more robust than the exact coefficient magnitudes, and starting-hero attribution gives the headline estimates a cleaner intention-to-treat interpretation than any rule that uses information revealed after the match has begun.

So this is not an oracle for the causal effect of pressing the hero-select button. It is a much better answer than raw win rate to a more carefully specified question: given two otherwise comparable match environments, what tends to happen when one team starts this hero rather than an average hero filling the same role? On 477,483 matches, the answer is not remotely the same for every hero.

Table 1: Hero adjusted plus-minus, relative to an average hero of the same role

	APM	S.E.	q	Starts	Pick (%)	Raw WR (%)	Play-time – Starting		APM	S.E.	q	Starts	Pick (%)	Raw WR (%)	Play-time – Starting
<i>Panel A: Tank (15 heroes)</i>								<i>Panel B: Damage (27 heroes)</i>							
The Thing	5.46***	(0.204)	0.000	122,883	2.12	52.4	+1.75	Black Cat	4.85***	(0.260)	0.000	66,494	1.15	55.2	+6.77
Magneto	1.96***	(0.139)	0.000	400,170	6.90	50.9	+1.60	Magik	3.47***	(0.219)	0.000	153,326	2.65	56.8	+11.11
Angela	1.87***	(0.261)	0.000	45,696	0.79	51.8	+0.81	Storm	3.09***	(0.305)	0.000	45,132	0.78	55.3	+7.18
Peni Parker	1.51***	(0.328)	0.000	43,799	0.76	54.6	+5.90	Daredevil	2.92***	(0.217)	0.000	75,418	1.30	53.6	+5.55
Hulk	1.33***	(0.219)	0.000	81,337	1.40	52.3	+2.21	Mister Fantastic	2.52***	(0.351)	0.000	40,940	0.71	50.4	+1.43
Thor	1.29***	(0.219)	0.000	80,452	1.39	50.4	-0.64	Iron Man	1.34***	(0.279)	0.000	50,391	0.87	51.5	+1.76
The Hood	0.45**	(0.172)	0.012	404,062	6.97	52.8	+6.08	Iron Fist	1.32***	(0.224)	0.000	85,109	1.47	52.1	+3.19
Devil Dinosaur	-0.03	(0.231)	0.907	69,629	1.20	48.6	-2.21	Hela	1.14***	(0.199)	0.000	110,771	1.91	49.8	+0.08
Groot	-0.67**	(0.290)	0.027	58,919	1.02	49.9	-3.13	Winter Soldier	1.00***	(0.141)	0.000	228,817	3.95	49.3	-0.83
Venom	-0.93***	(0.219)	0.000	103,844	1.79	50.2	-0.41	Wolverine	0.82**	(0.317)	0.012	42,210	0.73	49.3	-0.61
Doctor Strange	-0.98***	(0.202)	0.000	135,405	2.34	48.4	-5.85	Spider-man	0.58***	(0.175)	0.001	142,393	2.46	50.9	+2.32
Captain America	-1.22***	(0.352)	0.001	33,294	0.57	50.4	-0.57	Moon Knight	0.10	(0.297)	0.792	54,845	0.95	49.0	-3.47
Emma Frost	-1.77***	(0.192)	0.000	167,701	2.89	46.9	-4.91	Black Widow	0.02	(0.200)	0.907	84,335	1.46	48.7	-1.26
Rogue	-3.41***	(0.344)	0.000	52,686	0.91	51.0	+1.64	Blade	-0.20	(0.317)	0.578	47,659	0.82	49.8	+1.28
Deadpool (Vanguard)	-4.86***	(0.252)	0.000	40,623	0.70	47.9	-2.28	Black Panther	-0.21	(0.239)	0.423	65,701	1.13	50.4	+3.08
<i>Panel C: Support (13 heroes)</i>								Human Torch	-0.28	(0.351)	0.468	55,560	0.96	54.1	+4.84
Mantis	6.76***	(0.181)	0.000	148,036	2.55	56.9	+6.74	Elsa Bloodstone	-0.39	(0.229)	0.108	89,462	1.54	50.1	+0.26
Rocket Raccoon	4.14***	(0.150)	0.000	198,679	3.43	52.5	+1.03	Psylocke	-0.58	(0.342)	0.106	46,440	0.80	50.3	+2.12
Ultron	3.59***	(0.297)	0.000	39,543	0.68	53.3	+3.17	Deadpool (Duelist)	-0.67***	(0.220)	0.003	57,041	0.98	51.5	+2.52
Adam Warlock	2.08***	(0.315)	0.000	40,028	0.69	51.5	+1.84	Namor	-0.82***	(0.243)	0.001	75,351	1.30	45.2	-6.75
Loki	1.18***	(0.268)	0.000	48,317	0.83	51.3	+0.66	Hawkeye	-1.22***	(0.341)	0.001	40,267	0.69	48.8	-1.03
Cloak & Dagger	-0.04	(0.170)	0.852	320,390	5.53	49.9	+0.42	The Punisher	-1.81***	(0.262)	0.000	75,297	1.30	46.2	-8.64
Gambit	-0.30	(0.210)	0.178	195,439	3.37	49.9	-0.34	Star-lord	-2.41***	(0.300)	0.000	51,580	0.89	49.7	-1.64
Invisible Woman	-1.38***	(0.151)	0.000	228,763	3.95	48.0	-4.01	Scarlet Witch	-2.49***	(0.414)	0.000	46,118	0.80	46.1	-3.43
Jeff The Land Shark	-1.75***	(0.233)	0.000	71,748	1.24	46.8	-3.27	Squirrel Girl	-3.78***	(0.350)	0.000	32,402	0.56	42.9	-12.83
Jubilee	-1.91***	(0.172)	0.000	290,670	5.02	50.5	+2.23	Cyclops	-4.04***	(0.195)	0.000	108,037	1.86	45.2	-3.61
Luna Snow	-3.01***	(0.167)	0.000	197,630	3.41	46.6	-4.24	Phoenix	-4.28***	(0.345)	0.000	45,826	0.79	44.0	-9.38
White Fox	-3.48***	(0.208)	0.000	99,670	1.72	46.1	-3.21								
Deadpool (Strategist)	-5.89***	(0.210)	0.000	59,610	1.03	45.5	-1.02								
Matches	477,483							Log-likelihood	-320,067.7						
Player-matches	5,729,796							Heroes	55						

Notes. All 55 heroes are listed. *APM* is in percentage points of win probability at a balanced match ($\partial p / \partial x = \beta / 4$): the change from fielding this hero at match start in place of an average hero of the same role. Standard errors in parentheses are HC1, and the 95% confidence interval is $APM \pm 1.96$ S.E. Stars and q denote Benjamini–Hochberg false-discovery-rate control across the 55-hero family: * $q < 0.10$, ** $q < 0.05$, *** $q < 0.01$. Of the 55 heroes, 10 are not distinguishable from their role average. *Starts* is the number of player-slots that began the match on this hero and *Pick %* its share of all starting slots. Both reflect the crawl’s per-hero leaderboard seeding, not population pick rates. *Raw WR* is the play-time-weighted raw win rate, an assumption-light benchmark that controls for nothing. *Play-time – Starting* (W1–W0) is how far play-time-weighted attribution inflates the estimate relative to starting-hero attribution, in points. Sample: 482,997 matches crawled, less 2,139 containing draws (`is.win` = 2), 19 with incomplete play time, and 3,356 below a 240-second forfeit floor. Four team-up contrasts were structurally empty and dropped. Appendix Table A3 repeats every hero under all three attribution rules.

Appendix

Table A1: Specification checks

Check	Result	Interpretation
Placebo (permuted heroes)	0.023 vs 0.881	97.4% of signal destroyed
Sum-to-zero normalisation	within-role	holds at machine precision
Rank agreement, raw win rates	$\rho = 0.801$	up from 0.671 before correction
Within-role reparameterisation	$\rho = 0.9999$	mean change 0.010 pp
Camp indexing	49.87% camp 0	absolute map side, no crawl selection
Temporal holdout (7 days)	log loss 0.6674	vs 0.6928 constant, slope 1.067

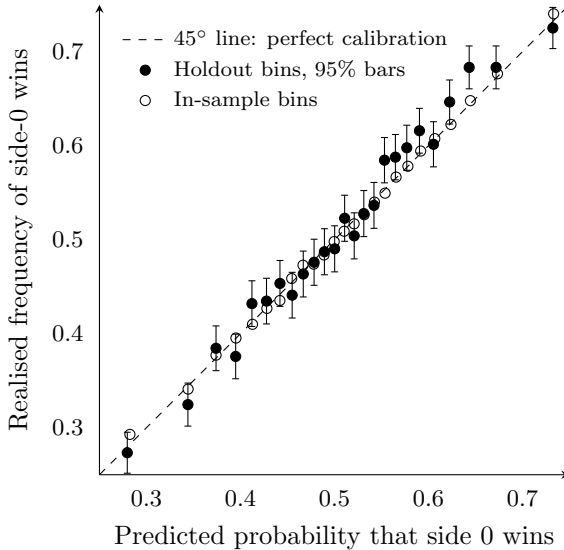
The placebo shuffles hero assignment against outcomes and refits. A placebo that failed would invalidate the specification. Camp-0 win rate varies 49.42%–52.52% across the 16 maps, which is why each map carries its own intercept. The holdout row summarises Figure A1.

Table A2: Out-of-sample fit as blocks of the model are added

Model	Parameters	Log-lik. (train)	Log loss (train)	Log loss (holdout)
Map intercepts only	16	−303,099	0.6929	0.6927
+ rank-score differential	17	−301,826	0.6900	0.6874
+ team-composition shapes	35	−299,416	0.6845	0.6819
+ hero contrasts	87	−294,223	0.6726	0.6698
+ team-up contrasts	187	−293,362	0.6706	0.6674

Each row adds one block to the model in the row above. The last row is the headline specification. Fitted on the 437,453 matches before the final 7 days, evaluated on the 40,030 matches of that week. A constant prediction at the training base rate gives a holdout log loss of 0.6928. The in-sample column shows how little of the improvement is overfitting.

Figure A1: Out-of-sample calibration of the headline model



Holdout, last 7 days, $N = 40,030$

Log loss	0.6674
constant prediction	0.6928
Brier score	0.2374
AUC	0.627
Calibration slope (s.e.)	1.067 (0.025)
Calibration intercept (s.e.)	0.018 (0.010)

Notes. The model is fitted on the 437,453 matches played before the final 7 days of the sample and evaluated on the 40,030 matches played during them, since a random split would leak meta drift. Matches are sorted by predicted probability into 25 equal-count bins of about 1,600 matches. Each filled circle is one bin's mean prediction against its realised win frequency, with a 95% binomial interval. Open circles repeat the construction in-sample on all 477,483 matches. The calibration slope and intercept are from a logit of the realised outcome on the log-odds of the prediction (ideal: 1 and 0). A constant prediction at the training base rate gives the log loss shown for comparison. Predicted probabilities fall between 5% and 93%, and 85% of holdout matches are predicted between 35% and 65%: the model separates unbalanced matches well and, as it should, treats balanced ones as coin flips.

Table A3: What happens to estimated hero strength when swaps affect attribution?

	<i>more post-start information →</i>					<i>more post-start information →</i>			
	Starting hero (W0)	Most-played hero (W2)	Play-time weighted (W1)	Difference (W1−W0)		Starting hero (W0)	Most-played hero (W2)	Play-time weighted (W1)	Difference (W1−W0)
<i>Panel A: Tank (15 heroes)</i>					<i>Panel B: Damage (27 heroes)</i>				
The Thing	5.46	4.94	7.21	+1.75	Black Cat	4.85	7.74	11.62	+6.77
Magneto	1.96	0.55	3.57	+1.60	Magik	3.47	7.94	14.57	+11.11
Angela	1.87	2.00	2.68	+0.81	Storm	3.09	6.57	10.27	+7.18
Peni Parker	1.51	10.18	7.41	+5.90	Daredevil	2.92	5.70	8.47	+5.55
Hulk	1.33	2.03	3.55	+2.21	Mister Fantastic	2.52	3.02	3.95	+1.43
Thor	1.29	0.73	0.65	-0.64	Iron Man	1.34	2.21	3.10	+1.76
The Hood	0.45	0.35	6.52	+6.08	Iron Fist	1.32	2.84	4.52	+3.19
Devil Dinosaur	-0.03	-0.56	-2.24	-2.21	Hela	1.14	1.04	1.22	+0.08
Groot	-0.67	-1.52	-3.79	-3.13	Winter Soldier	1.00	-0.07	0.17	-0.83
Venom	-0.93	-1.25	-1.35	-0.41	Wolverine	0.82	0.62	0.21	-0.61
Doctor Strange	-0.98	-2.55	-6.82	-5.85	Spider-man	0.58	1.48	2.90	+2.32
Captain America	-1.22	-0.51	-1.78	-0.57	Moon Knight	0.10	-2.01	-3.37	-3.47
Emma Frost	-1.77	-4.62	-6.68	-4.91	Black Widow	0.02	-0.96	-1.24	-1.26
Rogue	-3.41	-3.98	-1.78	+1.64	Blade	-0.20	0.28	1.09	+1.28
Deadpool (Vanguard)	-4.86	-5.79	-7.15	-2.28	Black Panther	-0.21	1.20	2.87	+3.08
<i>Panel C: Support (13 heroes)</i>					Human Torch	-0.28	1.74	4.56	+4.84
Mantis	6.76	9.88	13.50	+6.74	Elsa Bloodstone	-0.39	-0.46	-0.12	+0.26
Rocket Raccoon	4.14	4.50	5.18	+1.03	Psylocke	-0.58	0.08	1.54	+2.12
Ultron	3.59	6.92	6.76	+3.17	Deadpool (Duelist)	-0.67	0.48	1.85	+2.52
Adam Warlock	2.08	4.11	3.93	+1.84	Namor	-0.82	-3.87	-7.57	-6.75
Loki	1.18	2.14	1.84	+0.66	Hawkeye	-1.22	-1.31	-2.25	-1.03
Cloak & Dagger	-0.04	-1.84	0.38	+0.42	The Punisher	-1.81	-5.38	-10.45	-8.64
Gambit	-0.30	-2.07	-0.64	-0.34	Star-lord	-2.41	-2.64	-4.05	-1.64
Invisible Woman	-1.38	-3.32	-5.39	-4.01	Scarlet Witch	-2.49	-4.65	-5.91	-3.43
Jeff The Land Shark	-1.75	-3.04	-5.01	-3.27	Squirrel Girl	-3.78	-8.62	-16.61	-12.83
Jubilee	-1.91	-1.94	0.32	+2.23	Cyclops	-4.04	-5.45	-7.65	-3.61
Luna Snow	-3.01	-5.00	-7.25	-4.24	Phoenix	-4.28	-7.54	-13.67	-9.38
White Fox	-3.48	-4.73	-6.69	-3.21					
Deadpool (Strategist)	-5.89	-5.62	-6.91	-1.02					
Log-likelihood	-320,122	-305,869	-297,648		Cross-hero s.d.	2.60	4.27	6.38	
Rank corr. with starting hero	—	0.937	0.890						

Notes. Starting-hero attribution (W0) assigns the match to the hero selected at the start. Most-played-hero attribution (W2) assigns it to whichever hero the player used longest. Play-time weighting (W1) splits attribution according to realised play time. All three specifications use the same 477,483 matches, and estimates are within-role, in percentage points. Moving from W0 toward W1 gives the regression progressively more information about what happened after the match began. The coefficients become much larger, with cross-hero dispersion rising by a factor of 2.5, while the broad hero ranking stays comparatively stable. The difference (W1−W0) is reported rather than a ratio because a ratio diverges for heroes whose starting-hero estimate is near zero. The better in-sample likelihood under W1, −297,648 against −320,122, should not be read as evidence that W1 identifies hero strength better. Of course it predicts the match better. Part of the match has been placed inside the regressors.

Table A4: Team-up synergies, all 100 pairs

Team-up	Premium	S.E.	Team-up	Premium	S.E.	Team-up	Premium	S.E.
Squirrel Girl × Iron Man	8.66	(0.965)	White Fox × Black Cat	1.77	(0.587)	Loki × Hela	0.45	(0.708)
Blade × Moon Knight	8.30	(0.851)	Black Panther × Storm	1.77	(0.996)	Hela × Namor	0.21	(0.552)
Rogue × Gambit	6.95	(0.472)	Hela × Venom	1.74	(0.552)	Gambit × Jubilee	0.21	(0.322)
Adam Warlock × Ultron	6.82	(0.768)	Thor × Hela	1.71	(0.552)	Mister Fantastic × The Thing	0.17	(0.773)
Iron Fist × White Fox	6.61	(0.498)	Venom × Phoenix	1.67	(0.596)	Luna Snow × Adam Warlock	0.16	(0.763)
Mantis × Star-lord	4.27	(0.503)	Mister Fantastic × Rocket Raccoon	1.62	(0.591)	Devil Dinosaur × The Punisher	0.10	(0.700)
The Punisher × Rocket Raccoon	4.13	(0.422)	Blade × Captain America	1.62	(0.981)	Namor × Luna Snow	0.09	(0.483)
Storm × Jeff The Land Shark	3.86	(0.667)	The Punisher × Daredevil	1.58	(0.850)	Spider-man × Venom	0.06	(0.424)
Human Torch × Jubilee	3.71	(0.469)	Hawkeye × Cloak & Dagger	1.58	(0.542)	Groot × Jeff The Land Shark	-0.02	(0.819)
Loki × Mantis	3.53	(0.667)	Ultron × Iron Man	1.54	(0.864)	Psylocke × Cloak & Dagger	-0.03	(0.514)
Doctor Strange × Hulk	3.47	(0.603)	Emma Frost × Mantis	1.50	(0.363)	Venom × Blade	-0.04	(0.750)
Magik × The Hood	3.36	(0.295)	Daredevil × Black Widow	1.48	(0.728)	Captain America × Thor	-0.06	(1.079)
Adam Warlock × Storm	3.28	(1.149)	Wolverine × Gambit	1.47	(0.581)	Moon Knight × Elsa Bloodstone	-0.06	(0.940)
Luna Snow × White Fox	3.19	(0.447)	Magik × Doctor Strange	1.45	(0.389)	Winter Soldier × The Punisher	-0.07	(0.538)
Peni Parker × Rocket Raccoon	3.05	(0.543)	Jeff The Land Shark × Venom	1.41	(0.600)	Magneto × Emma Frost	-0.12	(0.313)
Cyclops × Phoenix	2.79	(0.641)	Angela × Star-lord	1.39	(1.006)	Black Cat × Black Panther	-0.25	(1.018)
Mantis × Adam Warlock	2.76	(0.831)	Angela × Loki	1.35	(0.957)	Jubilee × Blade	-0.41	(0.519)
Cloak & Dagger × The Hood	2.60	(0.233)	Invisible Woman × Human Torch	1.32	(0.507)	Invisible Woman × Mister Fantastic	-0.44	(0.614)
Doctor Strange × Invisible Woman	2.58	(0.326)	Gambit × Magneto	1.31	(0.277)	The Thing × Human Torch	-0.45	(0.548)
Scarlet Witch × Jubilee	2.47	(0.510)	Star-lord × Groot	1.24	(0.784)	Psylocke × Emma Frost	-0.49	(0.615)
Jubilee × The Hood	2.46	(0.236)	Elsa Bloodstone × Devil Dinosaur	1.21	(0.583)	Magneto × Scarlet Witch	-0.55	(0.511)
Black Widow × Phoenix	2.44	(0.859)	The Thing × Invisible Woman	1.20	(0.358)	Rocket Raccoon × Groot	-0.55	(0.518)
Hulk × Captain America	2.39	(0.998)	Rogue × Magneto	1.07	(0.501)	Thor × Angela	-0.81	(1.007)
Hawkeye × Psylocke	2.35	(0.907)	Cloak & Dagger × Luna Snow	1.07	(0.318)	Iron Man × Hulk	-0.86	(0.796)
Squirrel Girl × Spider-man	2.17	(0.959)	Captain America × Winter Soldier	0.98	(0.672)	Cyclops × Gambit	-1.00	(0.411)
Ultron × Peni Parker	2.11	(1.072)	Namor × Hulk	0.86	(0.585)	Storm × Thor	-1.17	(0.733)
Spider-man × Peni Parker	2.02	(0.714)	Devil Dinosaur × Jeff The Land Shark	0.83	(0.679)	Hulk × Wolverine	-1.39	(0.805)
Winter Soldier × Elsa Bloodstone	2.00	(0.442)	Phoenix × Rogue	0.79	(1.066)	Black Widow × Hawkeye	-1.54	(1.126)
Phoenix × Hela	1.98	(0.808)	Iron Fist × The Thing	0.73	(0.490)	Daredevil × Iron Fist	-1.80	(0.741)
Moon Knight × Cloak & Dagger	1.97	(0.465)	Emma Frost × Luna Snow	0.70	(0.332)	Human Torch × Storm	-1.80	(1.057)
Star-lord × Adam Warlock	1.95	(1.190)	Wolverine × Cyclops	0.68	(0.881)	Black Panther × Magik	-2.09	(0.666)
Iron Man × Thor	1.87	(0.688)	White Fox × Psylocke	0.61	(0.778)	Black Cat × Spider-man	-2.24	(0.627)
Scarlet Witch × Doctor Strange	1.85	(0.628)	Peni Parker × Black Panther	0.52	(1.041)			
Groot × Mantis	1.84	(0.535)	Rocket Raccoon × Squirrel Girl	0.50	(0.726)			

Notes. All 100 team-ups are listed in descending order of premium, reading down each column in turn. Each is labelled by its member heroes, anchor first. The premium is what a pair earns *beyond* what its two heroes contribute individually, in percentage points, under starting-hero attribution, with HC1 standard errors in parentheses. Four further team-ups are absent because they are defined against hero id 1057, base “Deadpool”, whose plays are all recorded under his three role variants, leaving those columns structurally empty.